AI FOR TRUSTWORTHY DECISION MAKING · STAGE 1

Federated activity recognition

Centralised training vs. FedAvg on the WISDM Actitracker phone-sensor data: baseline and a minimal federated setup.

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Motion data is personal.

A phone’s accelerometer is enough to tell walking from sitting, and it is in your pocket all day.

Federated learning trains one shared model while the raw data stays on each client. Only model weights travel to the server, which averages them (**FedAvg**).

RQ1 How close does FedAvg with 5 clients get to centralised training on the same data?

RQ2 Do more data and better features help, centralised and federated?

RQ3 How well do the models work for **new users**, and for every client equally?

Real phones, real people.

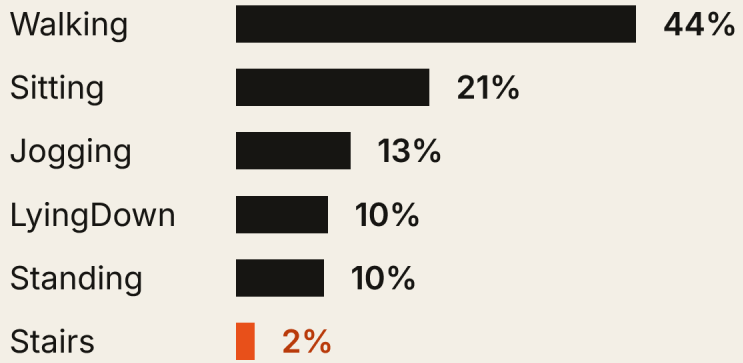
225
people (223 with complete windows)

2.98M
accelerometer readings at 20 Hz

13,381
windows of 10 seconds

6
activities, labelled by the users

SHARE OF WINDOWS PER ACTIVITY



Very imbalanced, so we report **balanced accuracy** and **macro-F1**, not only accuracy.

From raw readings to 10-second windows.

01

Clean

Drop broken lines and impossible values above 80 m/s².

02

Window

200 readings = 10 s. A window never mixes users, activities or gaps over 1 s.

03

Summarise

Mean, std, min, max per signal: 12 or 32 features. Notebook C: 71.

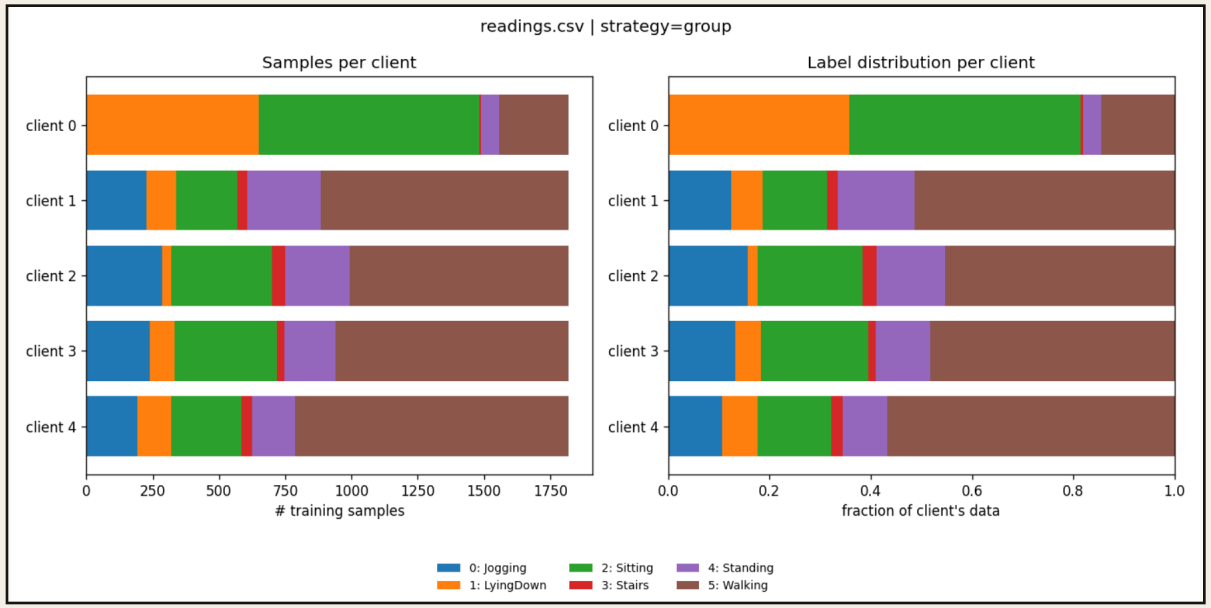
04

Hold out people

56 users (20 % of windows) are never seen in training: the test set is **new people**.

Four data variants (basic · rich · overlap · rich_overlap), all tested on the same 2,680 windows.

Five clients. One person, one client.



Client 0 is mostly Sitting and LyingDown; clients 1–4 are mostly Walking.

PARTITIONING

By person: non-IID

AGGREGATION

FedAvg, weighted by data size

TRAINING

10 rounds × 3 local epochs

MODEL

MLP: features → 256 → 128 → 6

FRAMEWORK

Flower simulation, seeds 1–3

Same accuracy. Not the same for everyone.

Overall: tied. Details: not.

TEST ACCURACY	BALANCED ACCURACY	WORST CLIENT
0.80	**0.68**	**0.92**
centralised	centralised	centralised
0.80	**0.61**	**0.55**
FedAvg · same	**FedAvg · −0.07**	**FedAvg · −0.37**

rich_overlap data, mean of 3 seeds. Test = 56 unseen users. Worst client = lowest per-client validation accuracy.

The cost comes from non-IID data, not from FL.

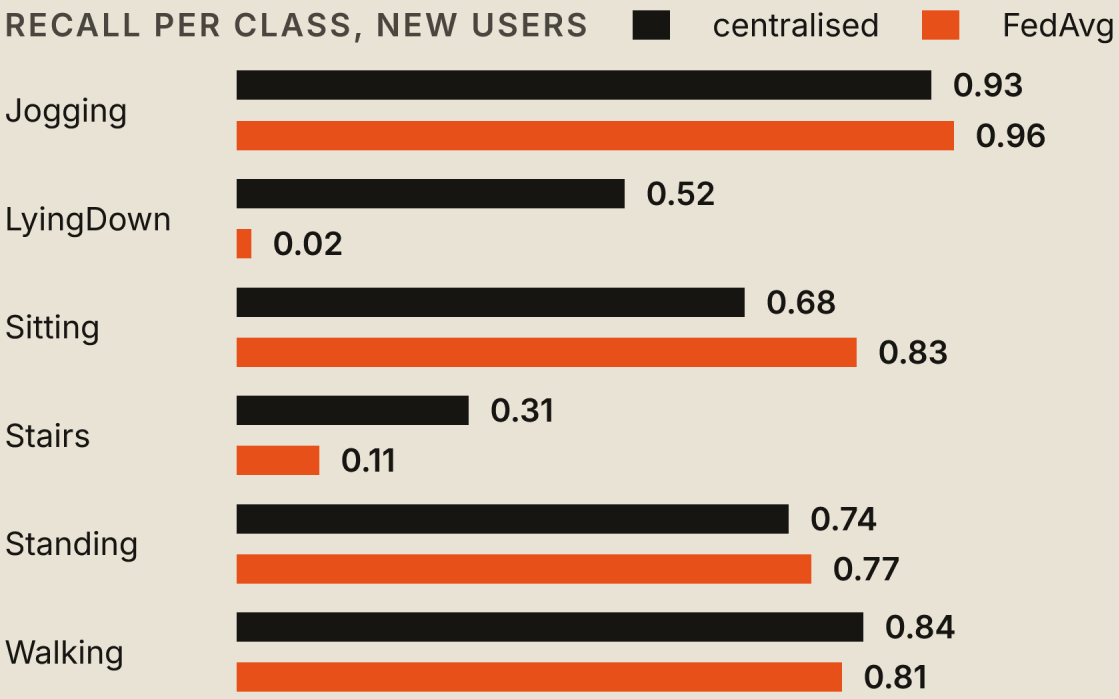
TEST BALANCED ACCURACY, BASIC DATA



Deal the same windows to 5 clients **at random** and FedAvg matches centralised.

Split them **by person** and it loses 7 points. Federated training itself costs almost nothing.

FedAvg almost never finds LyingDown.



125/130

LyingDown windows that FedAvg labels as **Sitting**. Stairs is mostly taken for Walking.

rich_overlap data, seed 1.

One client is left behind.

0.55

Client 0's accuracy with FedAvg, down from **0.92** with the centralised model.

Client	Most common	Central.	FedAvg
0	Sitting	0.92	0.55
1	Walking	0.96	0.87
2	Walking	0.93	0.80
3	Walking	0.95	0.89
4	Walking	0.97	0.94

Validation accuracy per client, rich_overlap. The averaged model fits the Walking-heavy majority.

Better features helped most.

TEST, NEW USERS	BASIC · 12	RICH · 32	HAND-CRAFTED · 71
Centralised, balanced acc.	0.64	0.67	0.73
FedAvg, balanced acc.	0.57	0.59	0.60
FedAvg, worst client	0.52	0.53	0.80

71 features add percentiles, rhythm and frequency bands. Rhythm is more alike between people than raw levels, so clients look less different.

Who is in the test set matters.

Work	Data	Test protocol	Accuracy
Kwapisz et al., 2011	WISDM lab, 36 people	10-fold CV over windows	91.7 %
Alsheikh et al., 2016	Actitracker	not user-independent; one-vs-rest “accuracy”*	98.2 %*
Peppas et al., 2020	Actitracker	test on Actitracker	79.1 %
Ours, centralised MLP	Actitracker, 71 features	56 unseen users	82.8 %
Ours, FedAvg (5 clients)	Actitracker, 71 features	56 unseen users	82.9 %

On **known** users our models reach 0.94, the same range as the papers. *Averages per-class (TP + TN) / all, so true negatives inflate it.

No setting fixes everything.

FedAvg

TEST ACCURACY
0.80

BALANCED ACC.
0.61

WORST CLIENT
0.55

FedProx, μ = 0.1

TEST ACCURACY
0.81 best

BALANCED ACC.
0.59

WORST CLIENT
0.60 best

FedAvg + balanced loss

TEST ACCURACY
0.75

BALANCED ACC.
0.66 best

WORST CLIENT
0.50

Overall accuracy, rare classes and the weakest client pull in different directions: measuring that trade-off is Stage 2.

What this baseline does not show yet.

Simple features

Mean/std/min/max ignore timing. Hand-crafted features or a 1D-CNN do better.

Data quality

Most LyingDown windows come from one person whose phone records almost no gravity.

Optimistic validation

Validation users are also training users. Only the test set measures new people.

Simulation

5 cross-silo clients. Real phones are thousands of small clients, often offline.

No privacy guarantee

The weights clients send can still leak information about their data.

Two trust dimensions next.

Fairness

WHAT WE SAW

Client 0 at 0.55; rare classes missed.

WHAT WE WILL TRY

FedProx (several μ), q-FedAvg, balanced loss.

HOW WE MEASURE IT

Worst-client accuracy, per-class recall.

Privacy

WHAT WE SAW

Model weights are shared unprotected.

WHAT WE WILL TRY

Differential privacy (DP-SGD noise; the repo reports ε).

HOW WE MEASURE IT

Accuracy vs. ε.

If time allows: robustness (FedMedian, Krum) and interpretability (feature importance).

TAKEAWAYS

Three things to remember.

1. FedAvg matches centralised accuracy on new users: **0.80 vs. 0.80**, without moving raw data.

2. The cost is in **rare activities** and the **minority client**, and it comes from non-IID data.

3. Better features close part of the gap. **Fairness** and **privacy** come next, in Stage 2.

Questions?